Ahmedur Rahman Shovon

Postdoctoral Appointee, Argonne National Laboratory

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Research Interests

High-Performance Computing	Runtime Systems, Parallel Programming Models, Agentic Workflows
Data Analytics	Declarative Analytics, Topological Data Analysis, Data Visualization
Performance Engineering	Performance Monitoring, Energy Efficiency, Cloud and Edge Systems

Education

PhD in Computer Science **2025**
University of Illinois Chicago *Chicago, IL*
Dissertation: Declarative Analytics on Heterogeneous HPC Systems
Advisor: Dr. Sidharth Kumar

MS in Information Technology **2018**
Jahangirnagar University *Dhaka, Bangladesh*
Thesis: A RESTful E-Governance Application Framework for People Identity Verification in Cloud
Advisor: Dr. Md Whaiduzzaman

BS in Information Technology **2016**
Jahangirnagar University *Dhaka, Bangladesh*
Project: EasyC: A Platform for Learning C Programming
Advisor: Dr. Risala Tasin Khan

Experience

Postdoctoral Appointee **08/2025 - p.**
Mathematics and Computer Science Division, Argonne National Laboratory *Lemont, IL*

Graduate Research and Teaching Assistant **08/2023 - 08/2025**
Department of Computer Science, University of Illinois Chicago *Chicago, IL*

Graduate Student Intern (WJ Cody Associate) **05/2023 - 08/2023**
Mathematics and Computer Science Division, Argonne National Laboratory *Lemont, IL*

Graduate Research Assistant (Blazer Fellow) **08/2021 - 08/2023**
Department of Computer Science, University of Alabama at Birmingham *Birmingham, AL*

Assistant Programmer **06/2019 - 08/2021**
ICT Division, Government of Bangladesh *Dhaka, Bangladesh*

Software Engineer **12/2017 - 06/2019**
Cefalo Bangladesh Ltd. *Dhaka, Bangladesh*

Software Engineer **10/2016 - 11/2017**
Codalo *Dhaka, Bangladesh*

Teaching Experience

Graduate Teaching Assistant **Spring 2025**
Department of Computer Science, University of Illinois Chicago *Chicago, IL*
Course: CS 455 Introduction to High-Performance Computing (class size: ∼60)

Graduate Teaching Assistant **Fall 2024**
Department of Computer Science, University of Illinois Chicago *Chicago, IL*
Course: CS 480 Database Systems (class size: ∼80)

Graduate Teaching Assistant **Fall 2016**
Institute of Information Technology, Jahangirnagar University *Dhaka, Bangladesh*
Course: IT 3107 Operating Systems (class size: ∼60)

Publications

Journal Articles (J)

[**J15**] Ahmedur Rahman Shovon, Sidharth Kumar, Gopikrishna Deshpande. **Topology Assisted Clustering of Temporal fMRI Brain Networks With Use-Case in Mitigating Non-Neural Multi-Site Variability**. *IEEE Access, 2025.* DOI: 10.1109/ACCESS.2025.3616256.

[**J14**] Sidharth Kumar, Ahmedur Rahman Shovon, Gopikrishna Deshpande. **The robustness of persistent homology of brain networks to data acquisition-related non-neural variability in resting state fMRI**. *Human Brain Mapping, 2023.* DOI: 10.1002/hbm.26403.

[**J13**] Darshan Shimoga Chandrashekar, Santhosh Kumar Karthikeyan, Praveen Kumar Korla, Henalben Patel, Ahmedur Rahman Shovon, Mohammad Athar, George J Netto, Zhaohui S Qin, Sidharth Kumar, Upender Manne, Chad J Crieghton, Sooryanarayana Varambally. **UALCAN: An update to the integrated cancer data analysis platform**. *Neoplasia, 2022.* DOI: 10.1016/j.neo.2022.01.001.

[**J12**] Md Whaiduzzaman, Md. Razon Hossain, Ahmedur Rahman Shovon, Shanto Roy, Aron Laszka, Rajkumar Buyya, Alistair Barros. **A Privacy-preserving Mobile and Fog Computing Framework to Trace and Prevent COVID-19 Community Transmission**. *IEEE Journal of Biomedical and Health Informatics (J-BHI), 2020.* DOI: 10.1109/JBHI.2020.3026060.

Book Chapters (B)

[**B11**] Ahmedur Rahman Shovon, Shanto Roy, Tanusree Sharma, Md. Whaiduzzaman. **A RESTful E-Governance Application Framework for People Identity Verification in Cloud**. *Luo M., Zhang LJ. (eds) Cloud Computing – CLOUD 2018. Lecture Notes in Computer Science, vol 10967. Springer, Cham.* DOI: 10.1007/978-3-319-94295-7_19.

Conference Papers (C)

[**C10**] Ahmedur Rahman Shovon, Yihao Sun, Thomas Gilray, Kristopher Micinski, Sidharth Kumar. **Multi-Node Multi-GPU Datalog**. *ACM International Conference on Supercomputing. ICS 2025.* DOI: 10.1145/3721145.3730431.

[**C9**] Yihao Sun, Ahmedur Rahman Shovon, Thomas Gilray, Kristopher Micinski, Sidharth Kumar. **Optimizing Datalog for the GPU**. *ACM International Conference on Architectural Support for Programming Languages and Operating Systems. ASPLOS 2025.* DOI: 10.1145/3669940.3707274.

[**C8**] Andres Sewell, Ke Fan, Ahmedur Rahman Shovon, Landon Dyken, Sidharth Kumar, Steve Petruzza. **Bruck Algorithm Performance Analysis for Multi-GPU All-to-All Communication**. *International Conference on High Performance Computing in Asia-Pacific Region. HPCAsia 2024.* DOI: 10.1145/3635035.3635047.

[**C7**] Ahmedur Rahman Shovon, Thomas Gilray, Kristopher Micinski, Sidharth Kumar. **Towards Iterative Relational Algebra on the GPU**. *USENIX Annual Technical Conference. USENIX ATC 2023.* ISBN: 978-1-939133-35-9.

[**C6**] Md Whaiduzzaman, Alistair Barros, Ahmedur Rahman Shovon, Md Razon Hossain, Colin Fidge. **A Resilient Fog-IoT Framework for Seamless Microservice Execution**. *IEEE International Conference on Services Computing. SCC 2021.* DOI: 10.1109/SCC53864.2021.00034.

[**C5**] Md Shahriare Satu, Md Khalilur Rahman, Maksud Alam Rony, Ahmedur Rahman Shovon, Md Jane Alam Adnan, Koushik Chandra Howlader, M Shamim Kaiser. **COVID-19: Update, Forecast and Assistant-An Interactive Web Portal to Provide Real-Time Information and Forecast COVID-19 Cases in Bangladesh**. *International Conference on Information and Communication Technology for Sustainable Development. ICICT4SD 2021.* DOI: 10.1109/ICICT4SD50815.2021.9396786.

[**C4**] Ahmedur Rahman Shovon, Shanto Roy, Arnab Kumar Shil, Tanjila Atik. **GDPR Compliance: Implementation Use Cases for User Data Privacy in News Media Industry**. *International Conference on Advances in Science, Engineering and Robotics Technology. ICASERT 2019.* DOI: 10.1109/ICASERT.2019.8934660.

[**C3**] Shanto Roy, Ahmedur Rahman Shovon, Md. Whaiduzzaman. **Combined approach of Tokenization and Mining to secure and optimize Big Data in Cloud Storage**. *IEEE R10 Humanitarian Technology Conference. R10-HTC 2017.* DOI: 10.1109/R10-HTC.2017.8288912.

Workshop Papers (W)

[**W5**] Ahmedur Rahman Shovon, Romain Pereira, Xingfu Wu, Brice Videau, Swann Perarnau. **Drava: An Event-Driven Runtime for Scientific Pipelines with Agentic Configuration Optimization**. *Workshop on Extreme-Scale Experiment-in-the-Loop Computing (XLOOP 2026), held with SC26.* (Accepted)

[**W4**] Ahmedur Rahman Shovon, Yihao Sun, Zhiling Lan, Swann Perarnau, Thomas Gilray, Kristopher Micinski, Michael E. Papka, Sidharth Kumar. **Heterogeneous Energy Characterization of GPU-Powered Datalog Engines**. *Workshop on Energy Efficiency with Sustainable Performance: Techniques, Tools, and Best Practices (EESP 2026), held with SC26.* (Accepted)

[**W3**] Ahmedur Rahman Shovon, Romain Pereira, Yihao Sun, Farhan Saif, Sidharth Kumar, Swann Perarnau. **Accelerating GPU Datalog with Kernel Fusion**. *Workshop on Irregular Applications: Architectures and Algorithms (IA3 2026), held with SC26.* (Accepted)

[**W2**] Kashyap Balakavi, Rushitha Janga, Ahmedur Rahman Shovon, Don Dempsey, Elliot Lefkowitz, Sidarth Kumar. **Scalable, interactive and hierarchical visualization of virus taxonomic data**. *IEEE Workshop on Visual Analytics in Healthcare (VAHC 2023), held with IEEE VIS 2023.* DOI: 10.1109/VAHC60858.2023.00012.

[**W1**] Ahmedur Rahman Shovon, Landon Richard Dyken, Oded Green, Thomas Gilray, Sidharth Kumar. **Accelerating Datalog applications with cuDF**. *Workshop on Irregular Applications: Architectures and Algorithms (IA3 2026), held with SC22.* DOI: 10.1109/IA356718.2022.00012.

Invited Talks and Presentations

Invited Talks

1. High-Performance Computing for Science in the Age of AI — Oct 2026
   International Conference on Machine Intelligence and Emerging Technologies (MIET 2026) — *Online*
2. Beyond the GPU — Sep 2026
   CS 594 Advanced Accelerator Programming, University of Illinois Chicago — *Chicago, IL*
3. Drava: An Event-Driven Runtime for Scientific Pipelines — Aug 2026
   MEERCAT Working Group — *Online*
4. Drava: An Event-Driven Runtime for Scientific Pipelines with Agentic Configuration Optimization — Jul 2026
   BIA Team, Argonne National Laboratory — *Lemont, IL*
5. Drava: An Event-Driven Runtime System for Scientific Workflows — Apr 2026
   Mathematics and Computer Science Division, Argonne National Laboratory — *Lemont, IL*
6. High Performance Joins — Feb 2024
   HPC Group, University of Illinois Chicago — *Chicago, IL*
7. Join on GPU — Oct 2023
   CS 480 Database Systems, University of Illinois Chicago — *Chicago, IL*
8. Porting CUDA Code to SYCL — Sep 2023
   oneAPI SYCL Hackathon — *Online*
9. LaTeXPlained — Aug 2022
   IEEE Student Branch, Jahangirnagar University, Bangladesh — *Online*
10. Graph Algorithms — Mar 2022
    CS 303 Algorithms and Data Structures, University of Alabama at Birmingham — *Birmingham, AL*

Poster Presentations

1. Drava: An Event-Driven Runtime for Scientific Pipelines with Agentic Configuration Optimization (Best Research Poster Nomination) — Nov 2026
   International Conference for High Performance Computing, Networking, Storage, and Analysis (SC26) *Chicago, IL*
2. Drava: An Event-Driven Runtime for Scientific Pipelines with Agentic Configuration Optimization — Oct 2026
   Postdoctoral Research and Career Symposium (PRCS 26), Argonne National Laboratory *Lemont, IL*
3. Scalable Multi-Node Multi-GPU Datalog Engine with Energy-Aware Profiling — Nov 2025
   International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25) *St. Louis, MO*
4. Scalable Multi-Node Multi-GPU Datalog Engine with Energy-Aware Profiling — Oct 2025
   Postdoctoral Research and Career Symposium (PRCS 25), Argonne National Laboratory *Lemont, IL*
5. MNMGDatalog: A Scalable Multi-Node Multi-GPU Datalog Engine — May 2025
   Greater Chicago Area Systems Research Workshop (GCASR 25) — *Chicago, IL*
6. MNMGDatalog: A Scalable Multi-Node Multi-GPU Datalog Engine (3rd Best Poster) — Apr 2025
   Computational Research Symposium 2025, Argonne National Laboratory — *Lemont, IL*
7. Accelerating Iterative Joins Toward a Modern Datalog Backend on GPU — May 2024
   Greater Chicago Area Systems Research Workshop (GCASR 24) — *Chicago, IL*

Awards

USENIX ATC Student Grant **2023**
Awarded by the USENIX Association.

Blazer Graduate Research Fellowship **2021-2022**
Awarded by the University of Alabama at Birmingham.

National Science and Technology (NST) Fellowship **2016-2017**
Awarded by the Ministry of Science and Technology, Bangladesh.

University Merit Scholarships **2012-2016**
Awarded by the Jahangirnagar University, Bangladesh.

Achievements

3rd @ Computational Research Symposium Poster Presentation 2025 **2025**
Presented poster on our multi-node multi-GPU Datalog engine (MNMGDatalog).

Winner @ oneAPI Hackathon: CUDA to SYCL Migration **2023**
Ported CUDA project to SYCL with automatic and manual interventions.

7th @ IEEEXtreme Programming Competition 9.0 **2015**
Achieved 7th position out of 2,477 teams, Team: *JUIITCoders*.

Technical Skills

Languages	Python, C++, JavaScript, Java, SQL, Bash, PHP
Parallel Computing	CUDA, MPI, SYCL, HIP, OpenMP, Thrust
Data Analytics	Pandas, NumPy, Scikit-learn, CuDF, Gudhi, Matplotlib, D3.js
Web and Cloud	Flask, Django, React, jQuery, CodeIgniter, Redis, AWS

Professional Service

Journal Reviewer

- IEEE Access
- Sustainable Cities and Society (Elsevier)

Conference Service

- Track Chair, MIET (AI/ML Track) 2026
- Technical Committee Member, IEEE Cloud Summit (AI/ML for Cloud and IoT Track) 2026
- Technical Committee Member, Communication Engineering and Cloud Computing Conference 2026
- Poster Reviewer, HPDC 2026
- Subreviewer, SC 2023, 2025

Proposal Reviewer

- DOE WDTS Visiting Faculty Program, Argonne National Laboratory 2026

Competition and Expo Judge

- DePaul University DemonHacks 2026
- University of Illinois Chicago SparkHacks 2026
- University of Illinois Chicago Undergraduate Research Forum 2024
- Research Expo, University of Alabama at Birmingham 2021–2023

Outreach and Leadership

- Member-at-large, Postdoctoral Society of Argonne, MCS Division, Argonne National Laboratory 2026
- Student Connect Facilitator, Argonne National Laboratory (13-student cohort) 2026
- Webmaster, IEEE Student Branch, Jahangirnagar University, Bangladesh 2015–2016